Practice sheet: decision trees, bagging and random forests

अभ्यास पत्र: decision trees, bagging और random forests

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How to use this sheet. This is for your own practice: you do not submit it. Try each problem first, then check the step-by-step solution. Every problem is asked in English and in Hindi; the solutions give the steps in English followed by a short Hindi explanation. Calculators are fine; keep four decimal places for entropy and Gini.
यह अभ्यास पत्र आपके अपने अभ्यास के लिए है, इसे जमा नहीं करना है। पहले हर problem खुद हल कीजिए, फिर step-by-step solution से मिलाइए। हर problem अंग्रेज़ी और हिंदी दोनों में दी गई है; solutions में steps अंग्रेज़ी में हैं और उसके बाद छोटा हिंदी स्पष्टीकरण है। Entropy और Gini के लिए चार दशमलव स्थान रखिए।

Formulas you need / ज़रूरी formulas

Entropy	$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$, where p is the fraction of class 1 in the group
Gini	$G(p) = 2p(1-p)$
Error rate	$E(p) = \min(p, 1-p)$
Gain of a split	impurity of parent $-\ \frac{n_L}{n}$ impurity$(L) - \frac{n_R}{n}$ impurity(R)
Pruning cost	$R_\alpha(T) = R(T) + \alpha\lvert T\rvert$ (R: training error rate, $\lvert T\rvert$: number of leaves)
Bootstrap sample	n draws with replacement; a given run is missed with probability $(1-\frac{1}{n})^n \to e^{-1} \approx 0.368$
Soft and hard vote	soft: average of the trees' leaf fractions; hard: share of trees whose leaf fraction exceeds $\frac{1}{2}$
Variance of an average	$\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\frac{\sigma^2}{B}$; floor as $B \to \infty$: $\rho\sigma^2$

Part A. Decision trees

Problem 1. Impurity of a group

A group has 6 spam emails and 2 not-spam emails. Find its entropy, Gini index and error rate. Repeat for a group with 3 spam and 3 not-spam.

एक group में 6 spam emails और 2 not-spam emails हैं। इसकी entropy, Gini index और error rate निकालिए। फिर 3 spam और 3 not-spam वाले group के लिए यही कीजिए।

Problem 2. Gain of one split

A training set has 10 emails: 6 spam and 4 not-spam. The question "contains a link?" sends 6 emails to the yes-group (5 spam, 1 not-spam) and 4 emails to the no-group (1 spam, 3 not-spam). Compute the gain using (a) entropy, (b) Gini, (c) error rate.

एक training set में 10 emails हैं: 6 spam और 4 not-spam। सवाल "क्या email में link है?" 6 emails को yes-group (5 spam, 1 not-spam) में और 4 emails को no-group (1 spam, 3 not-spam) में भेजता है। (a) entropy, (b) Gini, (c) error rate से gain निकालिए।

Problem 3. Which question first?

A training set has 12 emails: 7 spam and 5 not-spam. Three candidate questions give these (spam, not-spam) counts:

question	yes-group	no-group
contains a link	(6, 1)	(1, 4)
has an attachment	(3, 3)	(4, 2)
sender unknown	(5, 1)	(2, 4)

(a) Compute the entropy gain of each question and choose the first question. (b) Compute the error-rate gain of each. What do you notice about "has an attachment"?

12 emails का training set है: 7 spam और 5 not-spam। तीन उम्मीदवार सवाल ऊपर की table में (spam, not-spam) counts देते हैं। (a) हर सवाल का entropy gain निकालिए और पहला सवाल चुनिए। (b) हर सवाल का error-rate gain निकालिए। "has an attachment" के बारे में आपको क्या दिखता है?

Problem 4. Second question, and a prediction

Continue Problem 3 with the question "contains a link" at the root. Its yes-group has 7 emails (6 spam, 1 not-spam). Two candidate questions inside this group give:

question	yes-group	no-group
sender unknown	(4, 0)	(2, 1)
has an attachment	(3, 1)	(3, 0)

(a) Find the entropy gain of each and choose the question. (b) A new email has a link and a known sender. Which leaf does it reach, and what does the tree predict (give the leaf's fraction of spam)?

Problem 3 को आगे बढ़ाइए: root पर सवाल "contains a link" है। इसके yes-group में 7 emails हैं (6 spam, 1 not-spam)। इस group के अंदर दो उम्मीदवार सवाल ऊपर की table में हैं। (a) हर सवाल का entropy gain निकालकर सवाल चुनिए। (b) एक नए email में link है और sender known है। वह किस leaf तक पहुँचता है और tree क्या predict करता है (leaf में spam का fraction बताइए)?

Problem 5. Why the error rate cannot grow a tree

A group has 6 diverged and 2 converged training runs. A split sends 4 diverged runs to one side (pure) and 2 diverged plus 2 converged runs to the other side. Find the gain using the error rate, entropy and Gini. Which of the three fail to reward this split, and why?

एक group में 6 diverged और 2 converged training runs हैं। एक split एक तरफ 4 diverged runs (pure) और दूसरी तरफ 2 diverged और 2 converged runs भेजता है। Error rate, entropy और Gini से gain निकालिए। इनमें से कौन इस split को reward नहीं करता, और क्यों?

Problem 6. Pruning a tree

Three trees were grown on 100 training runs: T_1 has 8 leaves and 2 training mistakes; T_2 has 5 leaves and 6 mistakes; T_3 has 2 leaves and 15 mistakes. (a) Compute $R_\alpha(T) = R(T) + \alpha|T|$ for $\alpha = 0,\ 0.01,\ 0.02,\ 0.05$ and say which tree has the smallest cost each time. (b) For which α does T_2 start to beat T_1? For which α does T_3 start to beat T_2?

100 training runs पर तीन trees बने: T_1 में 8 leaves और 2 training गलतियाँ; T_2 में 5 leaves और 6 गलतियाँ; T_3 में 2 leaves और 15 गलतियाँ। (a) $\alpha = 0,\ 0.01,\ 0.02,\ 0.05$ के लिए $R_\alpha(T) = R(T) + \alpha|T|$ निकालिए और बताइए हर बार किस tree की cost सबसे कम है। (b) किस α से T_2, T_1 से बेहतर हो जाता है? किस α से T_3, T_2 से बेहतर हो जाता है?

Part B. Bootstrap and bagging / Bootstrap और bagging

Problem 7. Which runs does a bootstrap sample miss?

(a) For $n = 12$, find the probability that a given run is *not* drawn in a bootstrap sample of 12 draws, and the expected number of distinct runs in the sample. (b) Compare with $n = 100$ and with $n \to \infty$. (c) Find the probability that run 3 is drawn exactly twice when $n = 12$.

(a) $n = 12$ के लिए 12 draws वाले bootstrap sample में किसी दिए हुए run के न चुने जाने की probability और sample में distinct runs की expected संख्या निकालिए। (b) $n = 100$ और $n \to \infty$ से तुलना कीजिए। (c) $n = 12$ में run 3 के ठीक दो बार चुने जाने की probability निकालिए।

Problem 8. Out-of-bag runs of one sample

A bootstrap sample of the 12 training runs is $S = (1, 1, 3, 4, 4, 6, 7, 7, 7, 9, 10, 12)$. (a) Which runs are out of bag for this tree? (b) How many distinct runs does the tree see? (c) Which run was drawn most often, and how many times?

12 training runs का एक bootstrap sample $S = (1, 1, 3, 4, 4, 6, 7, 7, 7, 9, 10, 12)$ है। (a) इस tree के लिए कौन से runs out of bag हैं? (b) tree कितने distinct runs देखता है? (c) कौन सा run सबसे ज़्यादा बार चुना गया और कितनी बार?

Problem 9. Soft vote and hard vote

Five bagged trees give the fraction of spam in the leaf that a new email reaches: 0.55, 0.52, 0.51, 0.05, 0.02. (a) What does soft voting predict? (b) What does hard voting predict? (c) The two disagree: explain why.

पाँच bagged trees एक नए email के leaf में spam का fraction देते हैं: 0.55, 0.52, 0.51, 0.05, 0.02। (a) Soft voting क्या predict करती है? (b) Hard voting क्या predict करती है? (c) दोनों अलग हैं: कारण समझाइए।

Background: out-of-bag (OOB) error. Read this first.

To know how well a model will do on new data, you normally keep some training data aside as a validation set. Bagging gives you a validation set for free. Each tree is grown on a bootstrap sample, and a bootstrap sample misses about 37% of the runs (Problem 7). The runs a tree never saw are that tree's *out-of-bag* (OOB) runs.

For one training run, collect only the trees that did *not* draw it, let them vote, and compare their vote with the run's true class. The fraction of training runs on which this vote is wrong is the **out-of-bag error**. It is an honest estimate of the error on new data, because no run is ever predicted by a tree that trained on it.

Where it is used. (i) To estimate the error of a random forest without a separate validation set. (ii) To choose settings such as the number of trees B and the number of features offered at each split: pick the setting with the smallest OOB error. (In scikit-learn: `RandomForestClassifier(oob_score=True)`.)

Done for you: email 1. Three trees did not draw email 1, and 2 of them vote spam. The majority says spam, and the true class is spam, so the out-of-bag prediction for email 1 is correct. You will do emails 2 to 8.

पृष्ठभूमि: out-of-bag (OOB) error। पहले इसे पढ़िए। किसी model का नए data पर प्रदर्शन जानने के लिए आम तौर पर training data का कुछ हिस्सा validation set के रूप में अलग रख लिया जाता है। Bagging में यह validation set मुफ़्त मिल जाता है। हर tree एक bootstrap sample पर उगता है, और bootstrap sample लगभग 37% runs को छोड़ देता है (Problem 7)। जिन runs को tree ने कभी नहीं देखा, वे उस tree के out-of-bag (OOB) runs हैं।

किसी एक training run के लिए केवल वे trees लीजिए जिन्होंने उसे नहीं चुना, उनसे vote कराइए, और उनके vote की तुलना run की असली class से कीजिए। जितने training runs पर यह vote गलत निकले, उनका fraction **out-of-bag error** है। यह नए data की error का ईमानदार अनुमान है, क्योंकि किसी भी run की prediction ऐसे tree से नहीं होती जिसने उस पर training की हो।

इसका उपयोग कहाँ होता है। (i) अलग validation set के बिना random forest की error का अनुमान लगाने में। (ii) settings चुनने में, जैसे trees की संख्या B और हर split पर दिखाए जाने वाले features की संख्या: जिस setting पर OOB error सबसे कम हो उसे चुनिए। (scikit-learn में: `RandomForestClassifier(oob_score=True)`।)

आपके लिए हल किया हुआ: email 1। तीन trees ने email 1 को नहीं चुना और उनमें से 2 spam vote करते हैं। Majority spam कहती है और असली class spam है, इसलिए email 1 की out-of-bag prediction सही है। Emails 2 से 8 आप कीजिए।

Problem 10. Out-of-bag error

A bagged model with many trees is checked on its 8 training emails. For each email, only the trees that did *not* draw it vote:

email	1	2	3	4	5	6	7	8
true class	spam	spam	not	not	spam	not	spam	not
trees that did not draw it	3	4	3	5	2	4	3	3
of these, voting spam	2	1	0	3	2	1	3	1

(a) Give the out-of-bag prediction for emails 2 to 8 (email 1 is done for you above; the majority of the voting trees decides). (b) Compute the out-of-bag error. (c) A fully grown single tree makes 0 mistakes on its own training emails. Why is that number not a good estimate of new-email error, while the out-of-bag error is?

कई trees वाला bagged model अपने 8 training emails पर जाँचा जाता है। हर email के लिए केवल वे trees vote करते हैं जिन्होंने उसे नहीं चुना (table ऊपर)। (a) Emails 2 से 8 की out-of-bag prediction बताइए (email 1 ऊपर हल किया हुआ है; voting करने वाले trees का majority तय करता है)। (b) Out-of-bag error निकालिए। (c) एक पूरी तरह उगा single tree अपने ही training emails पर 0 गलतियाँ करता है। यह संख्या नए emails की error का अच्छा अनुमान क्यों नहीं है, जबकि out-of-bag error है?

Part C. Variance of an average / औसत का variance

Background: where σ^2 and ρ come from. Read this first.

Problems 11 to 15 use two numbers: σ^2 (how much one tree's answer varies) and ρ (how much two trees' answers move together). Both are measured at one *fixed test input*, by repeating the whole procedure many times.

The measurement. Repeat $K = 6$ times: draw a fresh training set, grow two trees on two different bootstrap samples, and write down their outputs T_1 and T_2 at the same test input. Here each tree predicts a real number (a regression tree), so σ^2 can be any positive number. (A classification tree that outputs a leaf fraction between 0 and 1 can never have σ^2 above 0.25.)

repeat	1	2	3	4	5	6
T_1	2.4	1.4	1.2	3.1	2.2	1.7
T_2	2.5	1.5	2.7	2.7	1.2	1.4
$T_1 - 2.0$	+0.4	−0.6	−0.8	+1.1	+0.2	−0.3
$T_2 - 2.0$	+0.5	−0.5	+0.7	+0.7	−0.8	−0.6
product	+0.20	+0.30	−0.56	+0.77	−0.16	+0.18

Done for you. Each column adds to 12.0, so each mean is 2.0 (the second and third rows are deviations from it).

Variance of one tree. Squares of the T_1 deviations: $0.16 + 0.36 + 0.64 + 1.21 + 0.04 + 0.09 = 2.50$; divide by $K - 1 = 5$: 0.50. For T_2: $0.25 + 0.25 + 0.49 + 0.49 + 0.64 + 0.36 = 2.48$, so 0.496. Both trees are grown the same way, so average them: $\sigma^2 \approx 0.498 \approx \mathbf{0.5}$.

Covariance and ρ. Sum of the products: $0.20 + 0.30 - 0.56 + 0.77 - 0.16 + 0.18 = 0.73$; divide by 5: covariance 0.146. Then $\rho = \dfrac{\text{covariance}}{\sigma^2} = \dfrac{0.146}{0.498} = 0.293 \approx \mathbf{0.3}$.

Check against the formula. The average of the two trees in each repeat is 2.45, 1.45, 1.95, 2.90, 1.70, 1.55 (mean 2.0); its variance is $\frac{1.61}{5} = 0.322$. The formula with $B = 2$ gives $\rho\sigma^2 + (1 - \rho)\sigma^2/2 = 0.146 + 0.176 = 0.322$. They agree.

In practice. With real data you have only one training set, so you cannot repeat the experiment on fresh training sets. σ^2 and ρ are then estimated by simulation, or by repeating the whole bootstrap procedure. In these problems the values $\sigma^2 = 0.5$ and $\rho = 0.3$ are simply given, and they are exactly what the measurement above produces.

Done for you: the variance of the average of $B = 4$ trees. Add the four trees and divide by 4: $\mathrm{Var}(\bar{T}) = \frac{1}{16}\mathrm{Var}(T_1 + T_2 + T_3 + T_4)$. The variance of a sum is the sum of all pairs: 4 terms $\mathrm{Var}(T_b) = \sigma^2 = 0.5$ and $4 \times 3 = 12$ terms $\mathrm{Cov}(T_b, T_c) = \rho\sigma^2 = 0.15$. So

$$\mathrm{Var}(\bar{T}) = \tfrac{1}{16}\,(4 \times 0.5 + 12 \times 0.15) = \tfrac{1}{16}(2.0 + 1.8) = \mathbf{0.2375}.$$

The shortcut formula agrees: $\rho\sigma^2 + (1-\rho)\sigma^2/B = 0.15 + 0.35/4 = 0.2375$. In Problem 11 you repeat this for other values of B.

पृष्ठभूमि: σ^2 और ρ कहाँ से आते हैं। पहले इसे पढ़िए। Problems 11 से 15 में दो संख्याएँ हैं: σ^2 (एक tree का उत्तर कितना बदलता है) और ρ (दो trees के उत्तर कितना साथ-साथ बदलते हैं)। दोनों एक fixed test input पर, पूरी प्रक्रिया को कई बार दोहराकर मापे जाते हैं।

मापन। $K = 6$ बार दोहराइए: नया training set लीजिए, दो अलग bootstrap samples पर दो trees उगाइए, और उसी test input पर उनके outputs T_1 और T_2 लिख लीजिए। यहाँ हर tree एक real संख्या predict करता है (regression tree), इसलिए σ^2 कोई भी धनात्मक संख्या हो सकती है। (जो classification tree 0 और 1 के बीच का leaf fraction देता है, उसका σ^2 कभी 0.25 से ऊपर नहीं जा सकता।) ऊपर की table में छह दोहराव हैं।

आपके लिए हल किया हुआ। हर column का जोड़ 12.0 है, इसलिए हर mean 2.0 है (दूसरी और तीसरी पंक्तियाँ उससे अंतर हैं)। एक tree का variance: T_1 के अंतरों के squares का जोड़ 2.50; $K - 1 = 5$ से भाग देने पर 0.50। T_2 के लिए 2.48, यानी 0.496। दोनों trees एक ही तरह उगाए गए हैं, इसलिए औसत लेने पर $\sigma^2 \approx 0.5$। Covariance और ρ: गुणनफलों का जोड़ 0.73; 5 से भाग देने पर covariance 0.146। फिर $\rho = 0.146/0.498 \approx 0.3$। जाँच: दो trees के औसत का variance सीधे निकालने पर $1.61/5 = 0.322$ आता है, और formula $\rho\sigma^2 + (1-\rho)\sigma^2/2 = 0.146 + 0.176 = 0.322$ भी वही देता है।

असल में। असली data में आपके पास एक ही training set होता है, इसलिए आप प्रयोग को नए training sets पर दोहरा नहीं सकते। तब σ^2 और ρ का अनुमान simulation से, या पूरी bootstrap प्रक्रिया दोहराकर, लगाया जाता है। इन problems में $\sigma^2 = 0.5$ और $\rho = 0.3$ सीधे दिए गए हैं, और ये ठीक वही हैं जो ऊपर का मापन देता है।

आपके लिए हल किया हुआ: $B = 4$ trees के औसत का variance। चार trees जोड़कर 4 से भाग दीजिए: $\mathrm{Var}(\bar{T}) = \frac{1}{16}\mathrm{Var}(T_1 + T_2 + T_3 + T_4)$। जोड़ का variance सभी जोड़ों का जोड़ है: 4 पद $\sigma^2 = 0.5$ वाले और $4 \times 3 = 12$ पद $\rho\sigma^2 = 0.15$ वाले। इसलिए $\frac{1}{16}(4 \times 0.5 + 12 \times 0.15) = \frac{1}{16}(2.0 + 1.8) = 0.2375$। Shortcut formula भी यही देता है: $0.15 + 0.35/4 = 0.2375$। Problem 11 में आप यही दूसरे B के लिए कीजिए।

Problem 11. Variance of the average

At a new input each tree has variance $\sigma^2 = 0.5$ and every pair of trees has correlation $\rho = 0.3$. The case $B = 4$ is done for you above. Find the variance of the average of (a) $B = 10$ trees, (b) $B = 100$ trees, (c) $B \to \infty$ trees. By what factor is the single-tree variance reduced in each case?

एक नए input पर हर tree का variance $\sigma^2 = 0.5$ है और हर दो trees के बीच correlation $\rho = 0.3$ है। $B = 4$ का मामला ऊपर हल किया हुआ है। (a) $B = 10$, (b) $B = 100$, (c) $B \to \infty$ trees के औसत का variance निकालिए। हर स्थिति में single-tree का variance कितने गुना घटा?

Problem 12. Independent against correlated trees

Take $\sigma^2 = 0.4$ and $B = 25$. Find the variance of the average when (a) $\rho = 0$, (b) $\rho = 0.5$, (c) $\rho = 1$. What does each case say about the use of averaging?

$\sigma^2 = 0.4$ और $B = 25$ लीजिए। (a) $\rho = 0$, (b) $\rho = 0.5$, (c) $\rho = 1$ के लिए औसत का variance निकालिए। हर स्थिति averaging के उपयोग के बारे में क्या बताती है?

Problem 13. How many trees?

With $\sigma^2 = 1$ and $\rho = 0.2$: (a) how many trees bring the variance of the average within 0.01 of its floor? (b) How many trees give a variance of at most 0.25? (c) Can the variance reach 0.15? Explain.

$\sigma^2 = 1$ और $\rho = 0.2$ के साथ: (a) कितने trees औसत के variance को उसकी floor के 0.01 के भीतर ले आते हैं? (b) कितने trees से variance अधिकतम 0.25 होता है? (c) क्या variance 0.15 तक पहुँच सकता है? कारण दीजिए।

Problem 14. Find ρ from measurements

In an experiment, the variance of a single tree's prediction at a test input is 0.12, and the variance of the average of $B = 10$ trees is 0.048. Find ρ and the floor $\rho\sigma^2$.

एक प्रयोग में एक test input पर single tree की prediction का variance 0.12 है और $B = 10$ trees के औसत का variance 0.048 है। ρ और floor $\rho\sigma^2$ निकालिए।

Problem 15. Estimate σ^2 from tree outputs

Six bagged trees give these leaf fractions at one test input (the box before Problem 11 shows how σ^2 is computed from a column of outputs): 0.9, 0.7, 0.8, 0.4, 1.0, 0.6. (a) What does the bagged model predict? (b) Estimate σ^2 using the sample variance (divide by $B - 1$). (c) State what σ^2 measures here, and why six numbers from fixed trees are enough to estimate it.

छह bagged trees एक test input पर ये leaf fractions देते हैं: $0.9,\ 0.7,\ 0.8,\ 0.4,\ 1.0,\ 0.6$। (a) Bagged model क्या predict करता है? (b) Sample variance ($B - 1$ से भाग देकर) से σ^2 का अनुमान लगाइए। (c) यहाँ σ^2 क्या मापता है, और fixed trees की छह संख्याएँ इसका अनुमान लगाने के लिए काफ़ी क्यों हैं?

Part D. Random forests

Problem 16. Which features does a split see?

A dataset has $p = 5$ features and a random forest offers $m = 2$ of them at each split. (a) How many different pairs can be offered? (b) What is the probability that the strongest feature is among the two offered? (c) What is the probability that two specific features are both offered? (d) With $p = 100$ and $m = 10$, what is the probability that the strongest feature is offered at a given split? (e) With the $m = 10$ setting and 15 splits in a tree, what is the probability that the strongest feature is never offered (treat splits as independent)?

एक dataset में $p = 5$ features हैं और random forest हर split पर उनमें से $m = 2$ दिखाता है। (a) कितने अलग-अलग जोड़े दिखाए जा सकते हैं? (b) सबसे मज़बूत feature दिखाए गए दो में होने की probability क्या है? (c) दो विशेष features दोनों के दिखाए जाने की probability क्या है? (d) $p = 100$ और $m = 10$ में किसी split पर सबसे मज़बूत feature दिखाए जाने की probability क्या है? (e) $m = 10$ के साथ और एक tree में 15 splits हों तो सबसे मज़बूत feature के कभी न दिखाए जाने की probability क्या है (splits को independent मानिए)?

Problem 17. The root question of a forest

In bagging, with all 3 features offered at every split, *rate* is the root question in 80.8% of the trees. In a random forest that offers 2 of 3 features at each split, *rate* wins in 88% of the trees in which it is offered. (a) In what fraction of trees is *rate* offered at the root? (b) In what fraction of trees is *rate* the root question? (c) Why is the answer to (b) smaller than 80.8%, and why does this lower the correlation ρ?

Bagging में, जब हर split पर सभी 3 features दिखाए जाते हैं, 80.8% trees में root का सवाल rate होता है। जिस random forest में हर split पर 3 में से 2 features दिखाए जाते हैं, उसमें rate जिन trees में दिखाया जाता है उनमें से 88% में जीतता है। (a) कितने fraction के trees में root पर rate दिखाया जाता है? (b) कितने fraction के trees में root का सवाल rate होता है? (c) (b) का उत्तर 80.8% से छोटा क्यों है, और इससे correlation ρ कम क्यों होता है?

Problem 18. Does the forest always win?

Bagging: $\sigma^2 = 0.071$, $\rho = 0.26$. Random forest: $\sigma^2 = 0.080$ (each tree is a little worse), $\rho = 0.20$. Use $B = 200$ trees. (a) Find the variance of the average for each. (b) Which is lower? (c) Suppose the forest's single-tree variance were 0.11 instead (with $\rho = 0.20$): does the forest still win?

Bagging: $\sigma^2 = 0.071$, $\rho = 0.26$। Random forest: $\sigma^2 = 0.080$ (हर tree थोड़ा कमज़ोर है), $\rho = 0.20$। $B = 200$ trees लीजिए। (a) हर एक के औसत का variance निकालिए। (b) कौन सा कम है? (c) मान लीजिए forest के single tree का variance 0.11 हो जाए ($\rho = 0.20$ के साथ): क्या forest तब भी जीतता है?

Step-by-step solutions

चरण-दर-चरण हल

Part A. Decision trees

Problem 1. Impurity of a group

Step 1: write p. Group 1: 6 spam of 8, so $p = \frac{6}{8} = 0.75$. Group 2: $p = \frac{3}{6} = 0.5$.

Step 2: apply each formula. Group 1: $H = -0.75\log_2 0.75 - 0.25\log_2 0.25 = 0.3113 + 0.5 = \mathbf{0.8113}$; $G = 2(0.75)(0.25) = \mathbf{0.375}$; $E = \min(0.75, 0.25) = \mathbf{0.25}$.
Group 2: $H = 1$, $G = 2(0.5)(0.5) = \mathbf{0.5}$, $E = \mathbf{0.5}$ (the largest value of each: a 50/50 group is the most mixed).

पहले p (spam का fraction) लिखिए, फिर तीनों formulas लगाइए। 50/50 group सबसे ज़्यादा mixed होता है, इसलिए तीनों की value वहीं सबसे बड़ी है।

Problem 2. Gain of one split

Step 1: parent impurity. $p = \frac{6}{10} = 0.6$: $H = 0.9710$, $G = 2(0.6)(0.4) = 0.48$, $E = 0.4$.

Step 2: children. Yes-group ($p = \frac{5}{6}$): $H = 0.6500$, $G = 2 \cdot \frac{5}{6} \cdot \frac{1}{6} = 0.2778$, $E = \frac{1}{6} = 0.1667$. No-group ($p = \frac{1}{4}$): $H = 0.8113$, $G = 2 \cdot \frac{1}{4} \cdot \frac{3}{4} = 0.375$, $E = 0.25$. Weights: $\frac{6}{10}$ and $\frac{4}{10}$.

Step 3: gain = parent − weighted children. (a) $0.9710 - 0.6(0.6500) - 0.4(0.8113) = 0.9710 - 0.3900 - 0.3245 = \mathbf{0.2564}$.
(b) $0.48 - 0.6(0.2778) - 0.4(0.375) = 0.48 - 0.1667 - 0.15 = \mathbf{0.1633}$.
(c) $0.4 - 0.6(0.1667) - 0.4(0.25) = 0.4 - 0.1 - 0.1 = \mathbf{0.2}$.

Gain = parent की impurity − दोनों children की impurity का weighted औसत (weights n_L/n और n_R/n)। तीनों measures में gain धनात्मक है, पर संख्याएँ अलग हैं।

Problem 3. Which question first?

Step 1: parent. $p = \frac{7}{12}$, so $H = 0.9799$.

Step 2: entropy of each child, then the gain.
Link: $H(\frac{6}{7}) = 0.5917$, $H(\frac{1}{5}) = 0.7219$. Gain $= 0.9799 - \frac{7}{12}(0.5917) - \frac{5}{12}(0.7219) = 0.9799 - 0.3452 - 0.3008 = \mathbf{0.3339}$.
Attachment: $H(\frac{1}{2}) = 1$, $H(\frac{4}{6}) = 0.9183$. Gain $= 0.9799 - \frac{6}{12}(1) - \frac{6}{12}(0.9183) = 0.9799 - 0.5 - 0.4591 = \mathbf{0.0207}$.
Sender unknown: $H(\frac{5}{6}) = 0.6500$, $H(\frac{2}{6}) = 0.9183$. Gain $= 0.9799 - 0.3250 - 0.4591 = \mathbf{0.1957}$.
(a) The largest gain is "contains a link" (0.3339): ask it first.

Step 3: error-rate gain. Parent error $\frac{5}{12} = 0.4167$. Link: $\frac{7}{12} \cdot \frac{1}{7} + \frac{5}{12} \cdot \frac{1}{5} = 0.0833 + 0.0833 = 0.1667$, gain **0.25**. Attachment: $\frac{6}{12} \cdot \frac{3}{6} + \frac{6}{12} \cdot \frac{2}{6} = 0.25 + 0.1667 = 0.4167$, gain **0**. Unknown: $\frac{6}{12} \cdot \frac{1}{6} + \frac{6}{12} \cdot \frac{2}{6} = 0.0833 + 0.1667 = 0.25$, gain **0.1667**.
(b) The error rate gives "has an attachment" a gain of exactly 0, although the question does change the class mix (50/50 and 67/33 groups); entropy gives it a small positive gain, 0.0207. The error rate cannot see the improvement.

हर सवाल के लिए parent की entropy में से दोनों children की weighted entropy घटाइए। सबसे बड़ा gain "link" का है, इसलिए वही पहला सवाल है। Error rate attachment को gain 0 देता है, जबकि entropy छोटा धनात्मक gain देती है।

Problem 4. Second question, and a prediction

Step 1: parent of this node. 6 spam, 1 not-spam: $H(\frac{6}{7}) = 0.5917$.

Step 2: gains. *Sender unknown*: groups (4,0), $H = 0$, and (2,1), $H(\frac{2}{3}) = 0.9183$. Gain $= 0.5917 - \frac{4}{7}(0) - \frac{3}{7}(0.9183) = 0.5917 - 0.3936 = \mathbf{0.1981}$.
Attachment: groups (3,1), $H(\frac{3}{4}) = 0.8113$, and (3,0), $H = 0$. Gain $= 0.5917 - \frac{4}{7}(0.8113) - 0 = 0.5917 - 0.4636 = \mathbf{0.1281}$.
(a) Choose "sender unknown" (0.1981).

Step 3: route the new email. It has a link (yes), and its sender is known, so "sender unknown" is no: it reaches the leaf with (2 spam, 1 not-spam).
(b) The leaf's spam fraction is $\frac{2}{3} = 0.667 > 0.5$: the tree predicts **spam**, with $\frac{2}{3}$ behind it. Only 3 training emails support this leaf.

पहले इस node की entropy निकालिए, फिर दोनों उम्मीदवारों का gain। नया email link → yes, unknown sender → no से (2 spam, 1 not-spam) वाले leaf में पहुँचता है और tree spam predict करता है (fraction 2/3)।

Problem 5. Why the error rate cannot grow a tree

Step 1: parent. $p = \frac{6}{8} = 0.75$: $H = 0.8113$, $G = 0.375$, $E = 0.25$.

Step 2: children, each with weight $\frac{4}{8}$. Pure side (4 diverged, 0 converged): $H = 0$, $G = 0$, $E = 0$. Mixed side (2, 2): $H = 1$, $G = 0.5$, $E = 0.5$.

Step 3: gain. Entropy: $0.8113 - \frac{4}{8}(0) - \frac{4}{8}(1) = \mathbf{0.3113}$. Gini: $0.375 - 0 - \frac{4}{8}(0.5) = \mathbf{0.125}$. Error rate: $0.25 - 0 - \frac{4}{8}(0.5) = \mathbf{0}$.
The error rate fails: it gives gain 0 to a split that clearly purifies one side. Reason: the error rate is concave but not *strictly* concave, so a chord can lie on the function and the weighted children can equal the parent. Entropy and Gini are strictly concave, so any split that changes the class mix lowers the weighted impurity.

Error rate में gain 0 आता है, जबकि entropy में 0.3113 और Gini में 0.125। Error rate strictly concave नहीं है, इसलिए वह एक तरफ़ pure करने वाले split को भी reward नहीं करता। Entropy और Gini strictly concave हैं, इसलिए दोनों reward करते हैं।

Problem 6. Pruning a tree

Step 1: training error rates. $R(T_1) = 0.02$, $R(T_2) = 0.06$, $R(T_3) = 0.15$; leaves $8, 5, 2$.

Step 2: costs $R + \alpha|T|$.

α	T_1 (8 leaves)	T_2 (5 leaves)	T_3 (2 leaves)
0	**0.02**	0.06	0.15
0.01	**0.10**	0.11	0.17
0.02	0.18	**0.16**	0.19
0.05	0.42	0.31	**0.25**

(a) The winners are T_1, T_1, T_2, T_3: a larger α prefers a smaller tree.

Step 3: crossing points. T_2 beats T_1 when $0.06 + 5\alpha < 0.02 + 8\alpha$, i.e. $3\alpha > 0.04$, so $\alpha > \mathbf{0.0133}$. T_3 beats T_2 when $0.15 + 2\alpha < 0.06 + 5\alpha$, i.e. $3\alpha > 0.09$, so $\alpha > \mathbf{0.03}$.

हर tree की cost $R + \alpha \times$leaves निकालिए। α बढ़ने पर छोटा tree जीतता है। दो trees की cost बराबर रखकर crossing point मिलता है: 0.0133 और 0.03।

Part B. Bootstrap and bagging

Problem 7. Which runs does a bootstrap sample miss?

(a) One draw misses a given run with probability $1 - \frac{1}{12} = \frac{11}{12}$; twelve independent draws all miss it with $(\frac{11}{12})^{12} = \mathbf{0.3520}$. So a run is drawn at least once with probability 0.6480 and the expected number of distinct runs is $12 \times 0.6480 = \mathbf{7.78}$.

(b) $n = 100$: $(0.99)^{100} = 0.3660$. $n \to \infty$: $e^{-1} = 0.3679$. About a third of the runs are always left out, and the fraction hardly depends on n.

(c) The number of times run 3 is drawn is binomial$(12, \frac{1}{12})$. $P(\text{exactly twice}) = \binom{12}{2}(\frac{1}{12})^2(\frac{11}{12})^{10} = 66 \times 0.006944 \times 0.4189 = \mathbf{0.192}$.

एक draw में run न चुने जाने की probability $11/12$ है; 12 draws में $(11/12)^{12} = 0.352$। इसलिए लगभग एक-तिहाई runs हर sample में छूट जाते हैं (out of bag)। ठीक दो बार चुने जाने के लिए binomial लगाइए: 0.192।

Problem 8. Out-of-bag runs of one sample

Step 1: list the distinct runs in S. $\{1, 3, 4, 6, 7, 9, 10, 12\}$.

Step 2: answers. (a) Out of bag: the runs from 1–12 not in this list: $\{\mathbf{2}, \mathbf{5}, \mathbf{8}, \mathbf{11}\}$. (b) The tree sees **8** distinct runs (the other 4 are out of bag, a fraction $\frac{4}{12} = 0.333$, close to the expected 0.352). (c) Run 7 appears three times (runs 1 and 4 twice each, the others once): $2+1+2+1+3+1+1+1 = 12$ draws, as it should be.

Sample में आए distinct runs की सूची बनाइए। बाकी runs out of bag हैं: $\{2, 5, 8, 11\}$। Tree 8 distinct runs देखता है; run 7 सबसे ज़्यादा (तीन) बार आया।

Problem 9. Soft vote and hard vote

(a) Soft. Average the leaf fractions: $\frac{1}{5}(0.55 + 0.52 + 0.51 + 0.05 + 0.02) = \frac{1.65}{5} = \mathbf{0.33} < 0.5$: predict **not spam**.

(b) Hard. Each tree votes spam if its fraction exceeds 0.5: trees 1, 2, 3 vote spam; trees 4, 5 vote not spam. Share for spam $= \frac{3}{5} = 0.6 > 0.5$: predict **spam**.

(c) Why they differ. Soft voting uses how confident each tree is. Three trees are barely above 0.5, while two are very sure the email is not spam (0.05, 0.02). The average gives those two confident trees a large pull. Hard voting throws away the confidence and counts every tree once, so the three borderline trees outvote the two confident ones. Neither rule is always right; soft voting is the usual default for probabilities.

Soft vote में fractions का औसत लेते हैं ($0.33 \to$ not spam); hard vote में हर tree का एक vote गिनते हैं ($3/5 \to$ spam)। Soft vote tree के confidence का इस्तेमाल करता है, hard vote नहीं करता, इसलिए दोनों अलग हो सकते हैं।

Problem 10. Out-of-bag error

Step 1: majority of the voting trees for each email. Spam if more than half of the voting trees say spam.

email	1	2	3	4	5	6	7	8
share voting spam	$\frac{2}{3}$	$\frac{1}{4}$	0	$\frac{3}{5}$	$\frac{2}{2}$	$\frac{1}{4}$	$\frac{3}{3}$	$\frac{1}{3}$
OOB prediction	spam	not	not	spam	spam	not	spam	not
true class	spam	spam	not	not	spam	not	spam	not
correct?	yes	**no**	yes	**no**	yes	yes	yes	yes

Step 2: error. (a) Predictions as in the table. (b) Out-of-bag error $= \frac{2}{8} = \mathbf{0.25}$ (emails 2 and 4).

(c) Why it is honest. A fully grown tree has seen its own training emails and memorises them, so 0 mistakes says nothing about new emails. For the out-of-bag prediction of an email, only trees that never saw that email vote, so each email acts as a fresh test email for those trees. No data needs to be set aside.

हर email के लिए केवल वे trees vote करते हैं जिन्होंने उसे नहीं देखा; majority से prediction बनती है। 8 में से 2 गलत हैं, इसलिए OOB error $= 0.25$। अकेला tree अपने training data को याद कर लेता है (0 error), पर OOB prediction में email उन trees के लिए नया होता है।

Part C. Variance of an average

Problem 11. Variance of the average

Step 1: formula. $\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$ with $\rho\sigma^2 = 0.3 \times 0.5 = 0.15$ and $(1-\rho)\sigma^2 = 0.35$.

Step 2: each B. (a) $B = 10$: $0.15 + 0.35/10 = \mathbf{0.185}$, a reduction of $0.5/0.185 = 2.7$ times. (b) $B = 100$: $0.15 + 0.0035 = \mathbf{0.1535}$, 3.3 times. (c) $B \to \infty$: **0.15**, 3.3 times at best $(0.5/0.15)$. More trees stop helping once $\mathrm{Var}(\bar{T})$ is near the floor set by ρ.

Formula $\rho\sigma^2 + (1-\rho)\sigma^2/B$ में संख्याएँ रखिए: 0.185, 0.1535, और floor 0.15। Floor तक पहुँचने के बाद और trees से लगभग कोई फ़ायदा नहीं होता।

Problem 12. Independent against correlated trees

Compute. (a) $\rho = 0$: $\sigma^2/B = 0.4/25 = \mathbf{0.016}$. (b) $\rho = 0.5$: $0.5(0.4) + 0.5(0.4)/25 = 0.2 + 0.008 = \mathbf{0.208}$. (c) $\rho = 1$: $0.4 + 0 = \mathbf{0.4}$.

Meaning. Independent trees (a) cut the variance by a factor $B = 25$. Moderately correlated trees (b) cut it only about $2\times$. Identical trees (c) gain nothing: averaging copies of the same tree is the same tree. Averaging helps only to the extent that the trees differ.

$\rho = 0$ में variance B गुना घटता है (0.016); $\rho = 0.5$ में सिर्फ़ लगभग 2 गुना (0.208); $\rho = 1$ में कुछ नहीं घटता (0.4)। Averaging तभी काम करती है जब trees एक-दूसरे से अलग हों।

Problem 13. How many trees?

Floor. $\rho\sigma^2 = 0.2$ and $(1-\rho)\sigma^2 = 0.8$.

(a) Within 0.01 of the floor: $0.8/B \le 0.01 \Rightarrow B \ge \mathbf{80}$.

(b) Variance ≤ 0.25: $0.2 + 0.8/B \le 0.25 \Rightarrow 0.8/B \le 0.05 \Rightarrow B \ge \mathbf{16}$.

(c) No. The variance is $0.2 + 0.8/B > 0.2$ for every finite B, so 0.15 lies below the floor and no number of trees reaches it. To go lower, ρ itself must fall (for example with a random forest).

Floor $= \rho\sigma^2 = 0.2$। (a) $0.8/B \le 0.01 \Rightarrow B \ge 80$; (b) $B \ge 16$; (c) नहीं: variance कभी 0.2 से नीचे नहीं जा सकता, उसके लिए ρ घटाना पड़ेगा।

Problem 14. Find ρ from measurements

Step 1: set up. $\sigma^2 = 0.12$, $B = 10$, so $\sigma^2/B = 0.012$. The formula gives $0.048 = 0.12\rho + 0.012(1-\rho) = 0.012 + 0.108\rho$.

Step 2: solve. $0.108\rho = 0.036$, so $\rho = \mathbf{0.333}$. Floor $= \rho\sigma^2 = 0.333 \times 0.12 = \mathbf{0.04}$.
Check: $0.04 + 0.667 \times 0.012 = 0.04 + 0.008 = 0.048$.

$0.048 = 0.12\rho + 0.012(1-\rho)$ को ρ के लिए हल कीजिए: $\rho = 1/3$, floor $= 0.04$। जाँच: $0.04 + 0.008 = 0.048$।

Problem 15. Estimate σ^2 from tree outputs

Step 1: mean. $\bar{T} = \frac{0.9+0.7+0.8+0.4+1.0+0.6}{6} = \frac{4.4}{6} = \mathbf{0.733}$. (a) $0.733 > 0.5$: the bagged model predicts **spam**.

Step 2: deviations. $+0.167$, -0.033, $+0.067$, -0.333, $+0.267$, -0.133. Squares: 0.0278, 0.0011, 0.0044, 0.1111, 0.0711, 0.0178; sum 0.2333.

Step 3: divide. (b) Sample variance $= 0.2333/5 = \mathbf{0.0467}$.

(c) Meaning. σ^2 is how much one tree's answer at this input varies from tree to tree; the six trees are

six draws from that variation, so their spread estimates it. The test input stays fixed; only the trees (their bootstrap samples) change.

पहले औसत ($0.733 \to$ spam), फिर हर tree का औसत से अंतर, उनके squares का जोड़ (0.2333), और $B-1=5$ से भाग: $\sigma^2 \approx 0.0467$। σ^2 एक fixed input पर trees के बीच का फैलाव है।

Part D. Random forests

Problem 16. Which features does a split see?

(a) $\binom{5}{2} = \mathbf{10}$ pairs.
(b) Pairs containing the strongest feature: choose its partner from the other 4, so 4 pairs: $\frac{4}{10} = \frac{m}{p} = \mathbf{0.4}$.
(c) Exactly one pair contains both specific features: $\frac{1}{10} = \mathbf{0.1}$.
(d) $\frac{m}{p} = \frac{10}{100} = \mathbf{0.1}$.
(e) Never offered in one split: 0.9; in 15 independent splits: $0.9^{15} = \mathbf{0.206}$. So about one tree in five never has the chance to use the strongest feature, which is exactly what makes forest trees differ.

सबसे मज़बूत feature किसी split पर दिखने की probability m/p है (0.4 और 0.1)। 15 splits में कभी न दिखने की probability $0.9^{15} = 0.206$: लगभग हर पाँचवें tree को यह feature मिलता ही नहीं।

Problem 17. The root question of a forest

(a) Rate is offered at the root in the pairs rate, batch and rate, warm-up: $\frac{2}{3}$ of the trees.
(b) Rate wins in 88% of those: $\frac{2}{3} \times 0.88 = \mathbf{0.587}$, i.e. 58.7% of the trees.
(c) In bagging, rate can always compete (and has two rivals), winning 80.8%. In the forest it is hidden from one third of the roots, so the other features get to start the tree: batch and warm-up start more trees (28.0% and 13.2% in the simulation). Trees that start with different questions grow differently and make different predictions, so their correlation ρ falls.

Rate root पर $2/3$ trees में दिखता है, और दिखने पर 88% जीतता है: $0.667 \times 0.88 = 0.587$। एक-तिहाई trees में वह छुपा रहता है, इसलिए दूसरे features से शुरू होने वाले trees बढ़ते हैं और ρ घटता है।

Problem 18. Does the forest always win?

(a) Bagging. Floor $= 0.26 \times 0.071 = 0.0185$; second term $0.74 \times 0.071/200 = 0.00026$; total $\mathbf{0.0187}$.

Forest. Floor $= 0.20 \times 0.080 = 0.0160$; second term $0.80 \times 0.080/200 = 0.00032$; total $\mathbf{0.0163}$.

(b) The forest is lower ($0.0163 < 0.0187$): the drop in ρ outweighs the small rise in σ^2.

(c) With $\sigma^2 = 0.11$: floor $0.20 \times 0.11 = 0.022$, second term $0.8 \times 0.11/200 = 0.00044$, total $\mathbf{0.0224} > 0.0187$. Now bagging wins: if hiding features makes each tree too weak, the forest loses. The number of features offered is therefore a setting to tune on held-out data (or with out-of-bag error).

Bagging: 0.0187; forest: 0.0163, इसलिए forest बेहतर है क्योंकि ρ घटा। पर अगर single tree का variance 0.11 हो जाए तो forest का variance 0.0224 है, यानी bagging बेहतर। इसलिए features की संख्या held-out data या OOB error से चुनिए।